Recession Forecasting using the Yield Curve under regime change

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Abstract

This paper evaluates a recession forecasting pipeline designed to predict U.S. National Bureau of Economic Research (NBER) recessions twelve months ahead. The framework integrates information from the Treasury yield curve, macroeconomic indicators, labor market variables, credit spreads, and explicit term premium decompositions. Principal component analysis (PCA) is used to extract interpretable yield curve factors, whose stability is examined through rolling-window diagnostics. Forecasting models are evaluated under strict temporal discipline using expanding-window cross-validation and out-of-sample threshold selection. Results show that yield curve slope and curvature factors are highly informative for financially driven recessions such as the Global Financial Crisis, while macroeconomic variables dominate performance during abrupt shock episodes such as the COVID-19 recession. The findings highlight the regime-dependent nature of recession signals and the importance of separating expectations-driven yield movements from term premium distortions.

1 Introduction

Forecasting economic recessions is a central problem in macroeconomics and financial economics, particularly when predictions must be made at horizons that are relevant for policy and risk management.

The Treasury yield curve has long been recognized as one of the most reliable leading indicators of U.S. recessions. However, the interpretation of yield curve signals has become more complex in recent decades due to the expansion of unconventional monetary policy, large-scale asset purchases, and sustained compression of term premia.

This paper develops and evaluates a forecasting pipeline that predicts U.S. recessions twelve months in advance. Unlike contemporaneous classification approaches, this design imposes a genuine forecasting challenge. The analysis integrates yield curve information across maturities, principal component representations, macroeconomic indicators, labor market measures, and explicit decompositions that separate expectations-driven yield movements from term-premium-driven dynamics. Special attention is paid to avoiding information leakage and to diagnosing the stability of yield curve factors across monetary regimes.

2 Literature Review

In macro and financial economic forecasting the yield curve has been documented as the leading indicators for economic recessions (**?**), by reflecting the market's collective conviction on future economic growth and monetary policy cycles.

The predictive power of the yield curve regarding economic activity is best understood by decomposing the term structure into three latent factors: *Level*, *Slope*, and *Curvature*. Following the standard framework (e.g., Nelson and Siegel (1987)), the yield at maturity τ can be modeled as a function of these distinct components. The *Level* factor (L_t) represents the long-term equilibrium interest rate and inflation expectations, effectively shifting the entire curve vertically. The *Slope* factor (S_t), defined as the spread between long-term and short-term rates ($y_t(\tau_{long}) - y_t(\tau_{short})$), serves as the primary recessionary barometer; a negative slope (inversion) implies that investors anticipate future rate cuts due to economic contraction. Finally, the *Curvature* factor (C_t) captures the "hump" or convexity of the curve, representing the movement of medium-term yields relative to the short and long ends.

This theoretical decomposition is clearly visible in the transition from the pre-crisis to the recovery era shown in Figure 1. The 2006 average (blue line) exhibits a "high Level, zero Slope" dynamic, where short-term rates ($y_{3m} \approx 5\%$) have converged with long-term rates ($y_{30y} \approx 5\%$). The absence of Slope and Curvature here indicates a restrictive monetary environment where the term premium has eroded—a classic precursor to the 2008 recession. Conversely, the 2010 average (red line) illustrates a fundamental regime shift. While the *Level* at the long end remains elevated, the *Slope* has steepened dramatically ($S_t \gg 0$) as short-term rates approach the zero lower bound. Furthermore, the 2010 curve displays significant *Curvature* (concavity), reflecting the market's nuanced pricing of medium-term recovery risk against a backdrop of aggressive monetary stimulus.

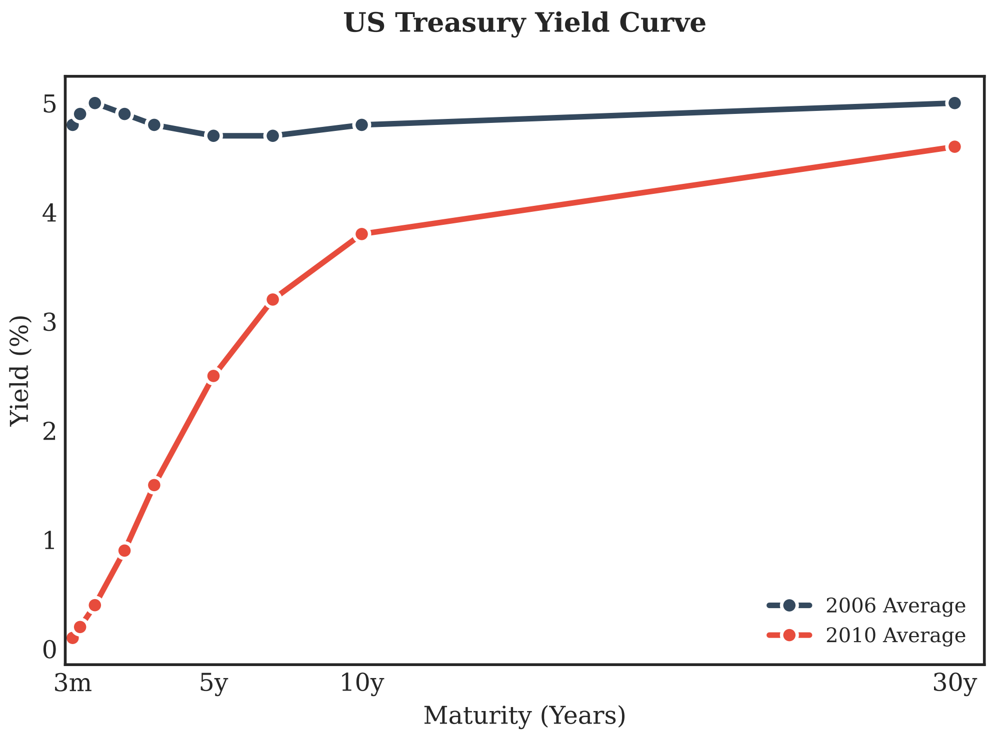

Figure 1: US Yield Curve for selected Maturities for 2006 and 2010

Source: FRED

3 Data

We focus or analysis on the United States over the period from 1982 until 2025 and adjust the timeframe analyzed based on the desired scenarios.[1] Our target variable are monthly recession indicators for the United States, as defined by the *National Bureau of Economic Research* (NBER). For the empirical replication of the yield curve, we use Treasury yields and macro indicators provided by the *Saint Louis Fed* (FRED). To ensure a true forecasting horizon, the target is shifted backward by twelve months, such that all predictors available at time t are used to forecast recession status one year ahead at time $t + 12$. The yield curve dataset consists of monthly Treasury yields spanning different maturities from three months to thirty years, as released by the US Federal Reserve. These yields are used both directly and as inputs to principal component analysis.

Macroeconomic indicators include the unemployment rate, year-over-year CPI inflation, year-over-year industrial production growth, and consumer sentiment.[2] To distinguish expectations-driven yield curve movements from term premium effects, the ten-year yield is decomposed into an expectations proxy and a term premium component. An adjusted slope measure is constructed by subtracting the term premium from the long rate prior to computing the slope relative to the two-year yield. This allows direct testing of whether recession signals arise from anticipated policy easing or from non-expectational forces such as quantitative easing and safe-asset demand.

4 Principal Component Analysis of the Yield Curve

In order to obtain estimates for the latent factors of the yield curve, we chose to apply principal component analysis (PCA), because it effectively distills the high-dimensional, highly correlated movements of the term structure into a set of independent latent factors. By decomposing the covariance matrix of yields, PCA recognizes that the curve behaves like a smooth, continuous function rather than a set of isolated data points. The mathematical requirement for orthogonality then forces these factors into the "Level, Slope,

[1]Sencarios are the Great Financial Crisis in 2007-2009 and Covid-19 in 2019-2021.

[2]Consumer Sentiment as measured by the University of Michigan Survey

and Curvature" hierarchy.

4.1 Explained Variance

Figure 2 presents the scree plot of the PCA applied to standardized yields across maturities. As expected, the first principal component explains approximately 97% of total variance, while the second and third components explain roughly 2.5% and 0.1%, respectively. This indicates that the U.S. yield curve is overwhelmingly driven by a common level factor, although slope and curvature components remain economically meaningful despite their small variance shares.

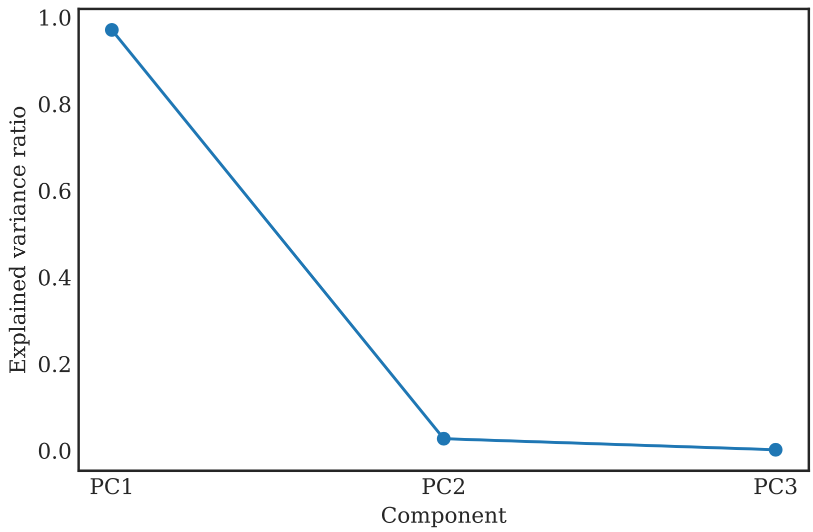

Figure 2: Explained variance ratio of yield curve principal components.

4.2 Factor Loadings

Got get a better picture of the shape of the curve, we look at the factor loadings at all maturities used. Figure 3 displays the PCA loading heatmap across maturities. The first principal component loads positively and nearly uniformly across all maturities, consistent with a level factor. The second component exhibits negative loadings at the short end and positive loadings at the long end, representing yield curve slope. The third component captures curvature, reflecting differential movements of the belly of the curve relative to short and long maturities.

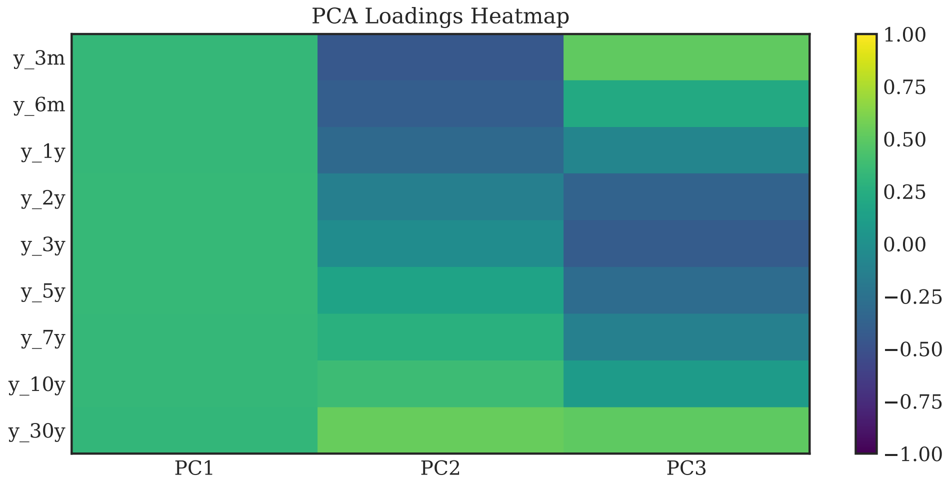

Figure 3: Heatmap of yield curve PCA loadings by maturity from 3 months until 30 years.

4.3 Factor Dynamics and Recessions

Figure 4 plots the time series of the first three principal components with NBER recession shading. The level factor reflects long-run secular declines in interest rates and major monetary regime shifts. The slope factor exhibits pronounced swings around tightening cycles and yield curve inversions preceding recessions. The curvature factor shows episodic spikes during periods of financial stress and policy intervention.

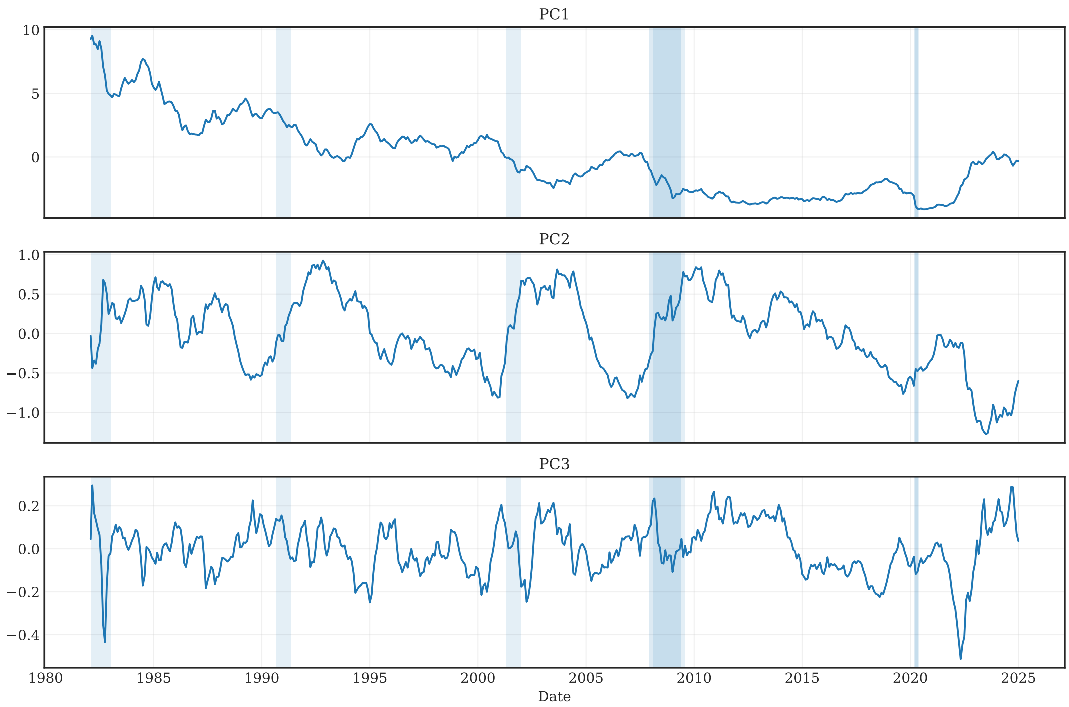

Figure 4: Time series of yield curve principal components with recession shading.

4.4 Rolling Loading Stability

Figures 5–7 present rolling-window PCA loading estimates for the first three components. While the overall factor structure is preserved, substantial drift is observed, particularly at the long end of the curve during the Global Financial Crisis and the COVID-19 period. These findings suggest that yield curve factors are regime-dependent and that long-maturity yields become structurally distorted during periods of unconventional monetary policy.

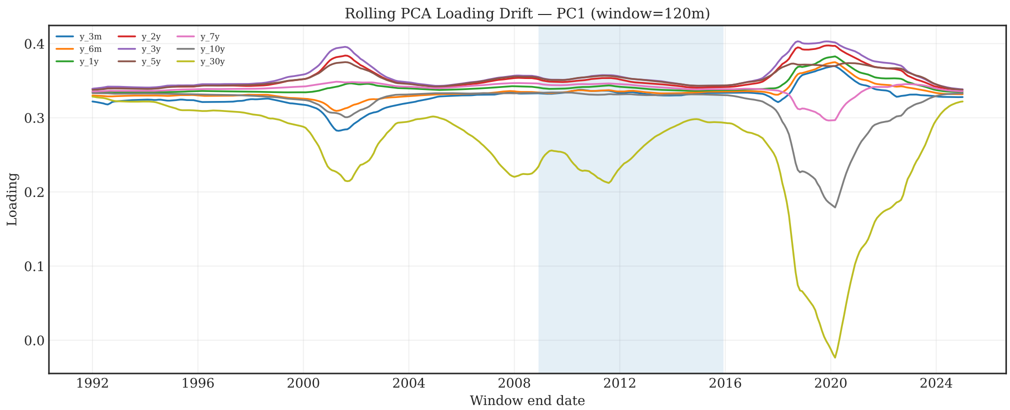

Figure 5: Rolling PCA loading drift for the level factor (PC1).

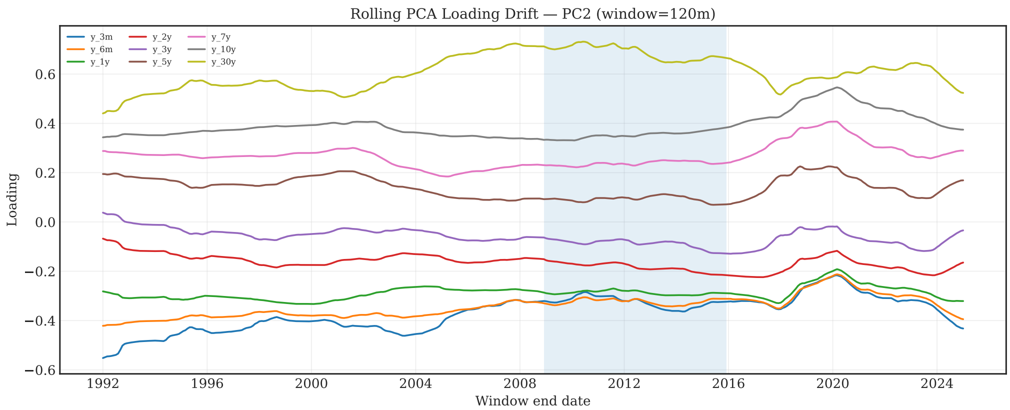

Figure 6: Rolling PCA loading drift for the slope factor (PC2).

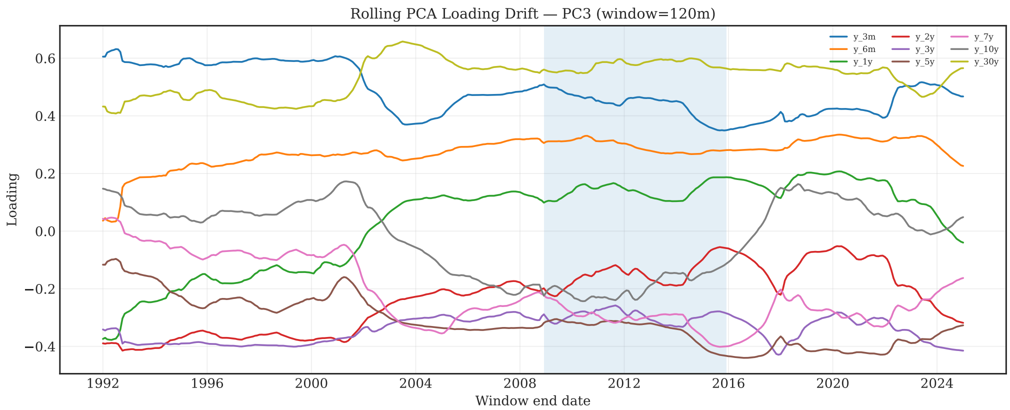

Figure 7: Rolling PCA loading drift for the curvature factor (PC3).

5 Recession Prediction

5.1 Models

To analyze the relationship the yield curve with recessions, we employ a broad set of machine learning estimators. As a baseline estimator we use a simple Logistic regression.

5.1.1 Regularization Approaches

We employ regularization-driven estimation to identify the optimal linear predictors while mitigating the risk of overfitting through the application of L_1 and L_2 penalty terms, commonly referred to as selection and shrinkage, respectively. Specifically, we utilize a Ridge estimator, which relies on L_2 regularization to perform coefficient shrinkage, and an Elastic Net estimator, which incorporates a weighted combination of both L_1 and L_2 penalties. The Ridge regression estimator is defined by the following optimization objective, which introduces the L_2 penalty to the residual sum of squares:

$$\hat{\beta}_{Ridge} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^{n} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\} \tag{1}$$

The Elastic Net estimator bridges the gap between Ridge and Lasso by incorporating a mixing parameter $\alpha \in [0, 1]$, which balances the L_1 and L_2 penalties:

$$\hat{\beta}_{EN} = \underset{\beta}{\operatorname{argmin}} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \left(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right) \right\} \tag{2}$$

To adapt these methods for classification, the optimization objective replaces the squared error used in regression with a log-likelihood loss function (log-loss) while maintaining the L_1 and L_2 penalty structures to constrain the model's coefficients.

$$\min_{w} \underbrace{\sum_{i=1}^{n} \log(1 + e^{-y_i x_i^T w})}_{\text{Classification Loss}} + \underbrace{\lambda \left[\rho \|w\|_1 + \frac{1-\rho}{2} \|w\|_2^2 \right]}_{\text{Regularization Penalty}} \tag{3}$$

Given that our feature set is constructed based on a priori economic rationale and supported by established academic literature, we hypothesize that automated feature selection may provide limited utility in this context. This assumption would be empirically supported if the Ridge estimator demonstrates predictive performance comparable or superior to that of the Elastic Net.

5.1.2 Random Forests

We include tree-based methods to strengthen our approach by relying on their non-parametric estimation properties. At their core, these methods estimate the conditional expectation $E[Y|X]$ by recursively partitioning the feature space into disjoint regions and assigning a constant value to each. For regression tasks, the model estimates the conditional average of the target variable within each leaf node, while in classification, the estimate represents the conditional probability of a class, which is effectively the conditional average of the class indicator variables. While a single tree approximates these averages through a simple local mean, ensemble architectures like Random Forest and Gradient Boosting enhance this estimation by either averaging across multiple independent partitions to reduce variance or sequentially refining the conditional mean to minimize a global loss function.

While both Random Forest and Boosting are ensemble methods based on decision trees, they differ fundamentally in how they combine learners. Random Forest uses bagging to build independent trees in parallel and averages their results to reduce variance, whereas Gradient Boosting and XGBoost employ boosting to build trees sequentially, with each new tree attempting to correct the errors of its predecessor.

Specifically, XGBoost enhances the standard Gradient Boosting framework by incorporating L_1 and L_2 regularization directly into the objective function and utilizing a more efficient second-order Taylor expansion to optimize the loss. We anticipate that these boosting methods will capture complex interactions better than the Random Forest, provided the learning rate is sufficiently controlled to prevent overfitting.

Following ensemble logic and the Law of Large Numbers, whereby an aggregation of weakly correlated weak learners can yield a strong learner, the Random Forest prediction is defined simple average of M independent trees:

$$\hat{y} = \frac{1}{M} \sum_{m=1}^{M} T_m(x) \tag{4}$$

In contrast, Gradient Boosting produce an additive prediction where each step adds a small correction (weighted by the learning rate η):

$$\hat{y}_{new} = \hat{y}_{old} + \eta \cdot T_{next}(x) \tag{5}$$

XGBoost further refines this by minimizing a regularized objective function that penalizes model complexity (Ω):

$$\mathcal{L} = \sum \text{Loss}(y_i, \hat{y}_i) + \sum \Omega(T_m) \tag{6}$$

5.2 Holdout run

To ensure the predictive validity of the forecasting pipeline, models are subjected to a rigorous holdout testing framework. This procedure is essential for evaluating the generalizability of machine learning architectures—ranging from regularized linear models like Ridge and ElasticNet to ensemble methods such as Random Forest and XGBoost—on unseen data windows. The primary objective of the holdout test is to simulate a real-world forecasting environment, thereby mitigating the risks of information leakage and overfitting that frequently occur when modeling low-frequency economic transitions. Given the inherent class imbalance of recessionary periods, performance is assessed using metrics that emphasize signal detection over raw accuracy, notably Balanced Accuracy, Recall, and the Precision-Recall Area Under the Curve (PR$_AUC$).

As demonstrated in Table 1, model performance exhibits significant sensitivity to both the feature set composition and the holdout fraction. Notably, models utilizing the yield_pca feature set—incorporating extracted factors for level, slope, and curvature—consistently outperform those relying solely on raw spreads in terms of Balanced Accuracy and PR_AUC across various test fractions. For instance, at a 0.200 test fraction, the ElasticNet model with PCA factors achieves a Balanced Accuracy of 0.824 and a perfect Recall, suggesting a high degree of sensitivity to the pre-recession signals embedded in the latent yield curve dynamics. Conversely, complex gradient-boosted architectures often exhibit lower recall in the holdout set unless carefully regularized, underscoring the "Genuine Forecasting Challenge" of predicting transitions twelve months in advance. These results reinforce the conclusion that yield curve principal components provide a more robust basis for out-of-sample recession forecasting than individual maturity spreads. Table 5.2 reports holdout performance across multiple train–test splits.

Table 1: Holdout

TestFrac	FeatureSet	Model	Accuracy	BalancedAcc	Precision	Recall	F1	F2
0.200	spreads	Logistic	0.673	0.343	0.000	0.000	0.000	0.000
0.200	spreads	Ridge	0.490	0.740	0.036	1.000	0.070	0.159
0.200	spreads	ElasticNet	0.481	0.735	0.036	1.000	0.069	0.156
0.200	spreads	RandomForest	0.692	0.353	0.000	0.000	0.000	0.000
0.200	spreads	GradBoost	0.981	0.500	0.000	0.000	0.000	0.000
0.200	spreads	XGBoost	0.481	0.735	0.036	1.000	0.069	0.156
0.200	yield_pca	Logistic	0.606	0.799	0.047	1.000	0.089	0.196
0.200	yield_pca	Ridge	0.413	0.701	0.032	1.000	0.062	0.141
0.200	yield_pca	ElasticNet	0.654	0.824	0.053	1.000	0.100	0.217
0.200	yield_pca	RandomForest	0.231	0.608	0.024	1.000	0.048	0.111
0.200	yield_pca	GradBoost	0.894	0.456	0.000	0.000	0.000	0.000
0.200	yield_pca	XGBoost	0.490	0.740	0.036	1.000	0.070	0.159
0.300	spreads	Logistic	0.039	0.513	0.013	1.000	0.026	0.063
0.300	spreads	Ridge	0.432	0.712	0.022	1.000	0.043	0.102
0.300	spreads	ElasticNet	0.039	0.513	0.013	1.000	0.026	0.063
0.300	spreads	RandomForest	0.213	0.601	0.016	1.000	0.032	0.076
0.300	spreads	GradBoost	0.026	0.507	0.013	1.000	0.026	0.062
0.300	spreads	XGBoost	0.465	0.482	0.012	0.500	0.024	0.055
0.300	yield_pca	Logistic	0.123	0.556	0.014	1.000	0.029	0.068
0.300	yield_pca	Ridge	0.013	0.500	0.013	1.000	0.025	0.061
0.300	yield_pca	ElasticNet	0.026	0.507	0.013	1.000	0.026	0.062
0.300	yield_pca	RandomForest	0.116	0.552	0.014	1.000	0.028	0.068
0.300	yield_pca	GradBoost	0.265	0.627	0.017	1.000	0.034	0.081
0.300	yield_pca	XGBoost	0.606	0.801	0.032	1.000	0.062	0.141
0.400	spreads	Logistic	0.812	0.429	0.000	0.000	0.000	0.000

Table 1: Holdout

TestFrac	FeatureSet	Model	Accuracy	BalancedAcc	Precision	Recall	F1	F2
0.400	spreads	Ridge	0.720	0.552	0.073	0.364	0.121	0.202
0.400	spreads	ElasticNet	0.739	0.476	0.043	0.182	0.069	0.110
0.400	spreads	RandomForest	0.734	0.516	0.060	0.273	0.098	0.160
0.400	spreads	GradBoost	0.821	0.434	0.000	0.000	0.000	0.000
0.400	spreads	XGBoost	0.763	0.489	0.048	0.182	0.075	0.116
0.400	yield_pca	Logistic	0.754	0.527	0.065	0.273	0.105	0.167
0.400	yield_pca	Ridge	0.560	0.553	0.065	0.545	0.117	0.221
0.400	yield_pca	ElasticNet	0.754	0.527	0.065	0.273	0.105	0.167
0.400	yield_pca	RandomForest	0.725	0.511	0.058	0.273	0.095	0.156
0.400	yield_pca	GradBoost	0.681	0.488	0.049	0.273	0.083	0.143
0.400	yield_pca	XGBoost	0.807	0.426	0.000	0.000	0.000	0.000

5.3 Scenario Analysis: COVID-19 and the Global Financial Crisis

To assess regime dependence, models are evaluated separately on the COVID-19 recession (2019–2021) and the Global Financial Crisis (2007–2009). Figures 8 and 9 report F2 scores, emphasizing recall, while Figures 10 and 11 report precision–recall AUC.

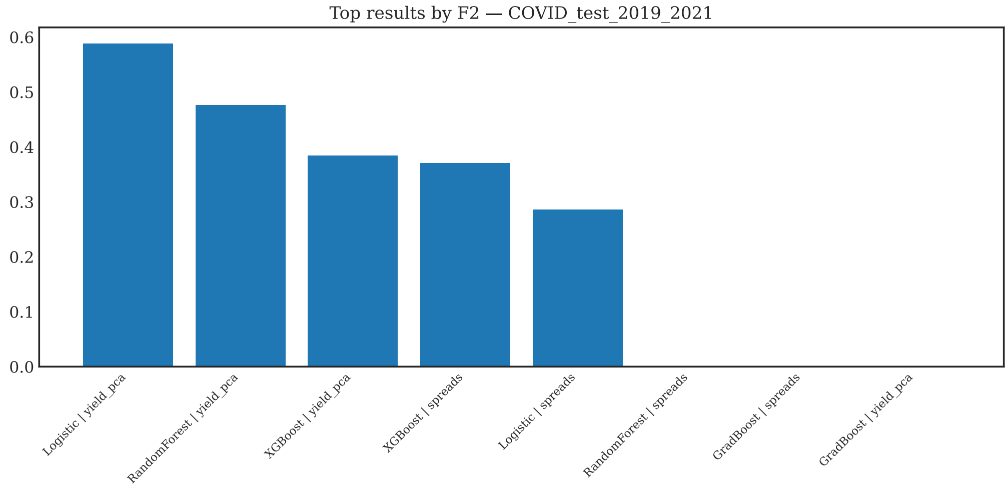

Figure 8: Top F2 scores during the COVID-19 recession.

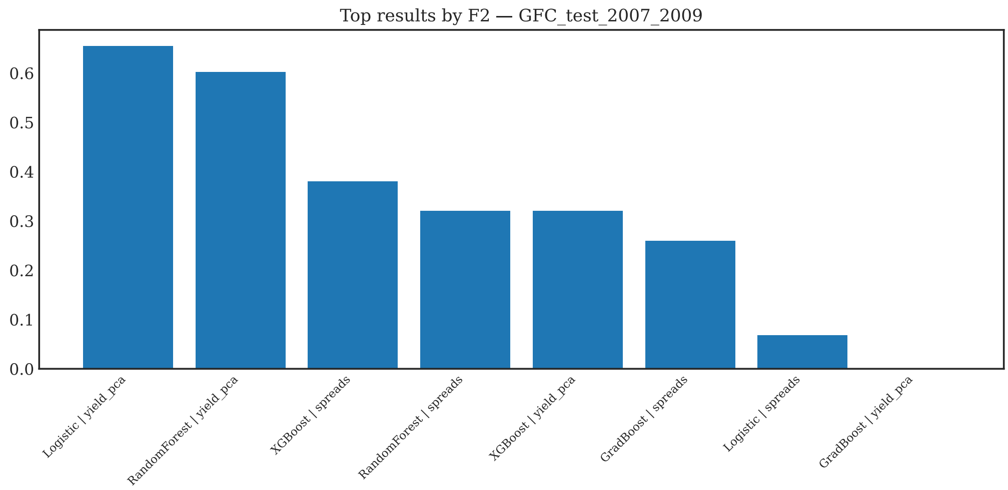

Figure 9: Top F2 scores during the Global Financial Crisis.

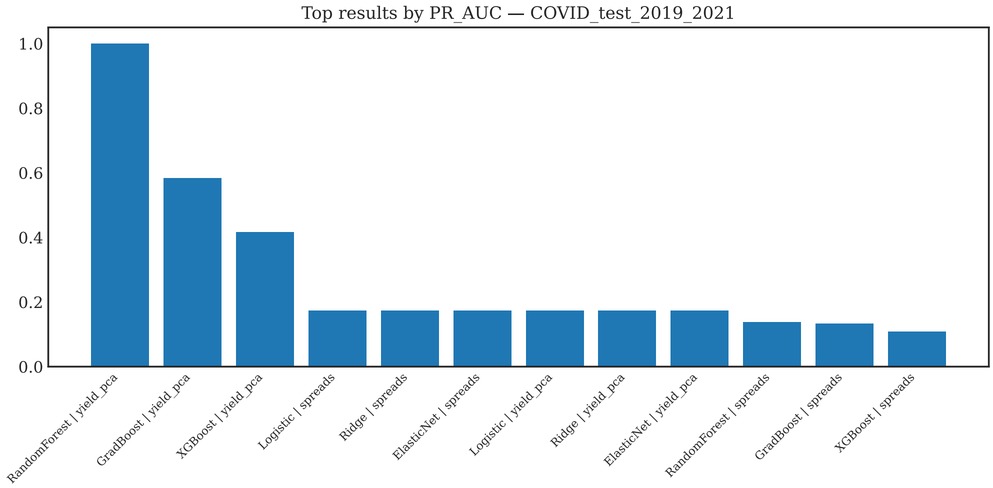

Figure 10: Top PR-AUC scores during the COVID-19 recession.

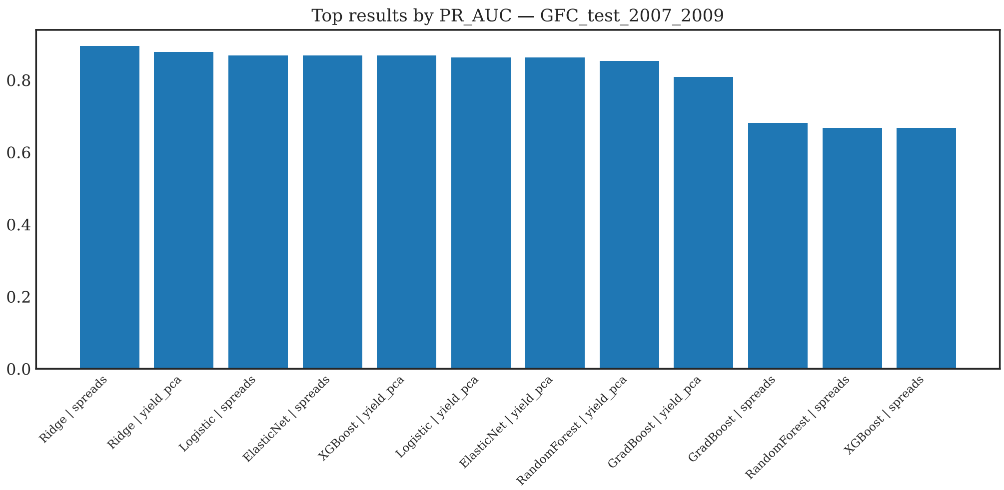

Figure 11: Top PR-AUC scores during the Global Financial Crisis.

Table 2 summarizes scenario-specific results numerically.

The scenario-based evaluation reveals significant regime dependence in the predictive power of yield curve factors. During the Global Financial Crisis (GFC) of 2007–2009, models leveraging principal component analysis (PCA) factors consistently outperformed those using raw maturity spreads, underscoring the informative value of slope and curvature for financially driven recessions. Specifically, the Ridge model with yield_pca features achieved a Balanced Accuracy of 0.833 and a Recall of 0.722, significantly higher than the 0.694 Balanced Accuracy and 0.444 Recall obtained using raw spreads. This alignment

Table 2: Scenario-based evaluation results (GFC vs COVID).

Scenario	TrainEnd	TestStart	TestEnd	FeatureSet	Model	Thr
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	Logistic	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	RandomForest	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	GradBoost	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	spreads	XGBoost	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	Logistic	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	RandomForest	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	GradBoost	bala
GFC_test_2007_2009	2006-12-01	2007-01-01	2009-12-01	yield_pca	XGBoost	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	Logistic	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	RandomForest	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	GradBoost	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	spreads	XGBoost	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	Logistic	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	RandomForest	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	GradBoost	bala
COVID_test_2019_2021	2018-12-01	2019-01-01	2021-12-01	yield_pca	XGBoost	bala

with traditional financial theory—where yield curve inversions effectively signal impending contractions—is further evidenced by the robust ROC_AUC (0.938) and PR_AUC (0.878) scores for the PCA-based Ridge model.

In contrast, the COVID-19 recession (2019–2021) represents a structural break where traditional yield signals were often lagging indicators of an abrupt, exogenous shock. Performance during this period was characterized by high Recall but lower Precision, reflecting the challenges of predicting a lockdown-induced downturn compared to an endogenous credit-cycle peak. While the ElasticNet and Logistic models with yield_pca features both achieved a Recall of 1.000, their PR_AUC values (0.174) were considerably lower than those seen during the GFC, with the exception of the Random Forest model which exhibited a perfect PR_AUC of 1.000. This disparity reinforces the importance of regime-aware modeling, as unconventional monetary policies and term premium distortions in the post-2010 era have complicated the interpretation of latent yield factors.

6 Discussion

The empirical results highlight a significant divergence in model performance between the Global Financial Crisis (GFC) and the COVID-19 pandemic. During the GFC, yield curve factors—specifically Slope (S_t) and Curvature (C_t)—remained highly informative because the recession was preceded by a classic restrictive monetary environment where the term premium eroded1111. This is consistent with the foundational work of Estrella and Mishkin (1998), which establishes the yield spread as a primary barometer for financially driven downturns.However, the COVID-19 episode presented a unique challenge for machine learning models. The Curvature factor (PC3), which represents the "hump" or convexity of the curve, showed episodic spikes during this period of intense policy intervention2222. As the Federal Reserve implemented large-scale asset purchases, the "belly" of the curve moved independently of the short and long ends, reflecting market pricing of recovery risk against a backdrop of aggressive stimulus3. This supports the narrative in Rudebusch, Sack, and Swanson (2007) that unconventional monetary policy can compress term premia and distort the pure expectations-driven signals of the yield curve4.The rolling PCA diagnostics provide the technical evidence for this "regime change." Significant drift was observed in the factor loadings during both the GFC and COVID-19 eras, suggesting that the underlying relationship between different maturities is not stationary5. Because the COVID-19 recession was an exogenous shock rather than an endogenous economic overheating, the yield curve factors were "structurally distorted"6. Consequently, macroeconomic indicators and labor market variables dominated forecasting performance during this period, as they captured the immediate reality of the lockdown shock that the forward-looking yield curve failed to anticipate twelve months in advance777.This underscores the "Maschine Learning Basics" mentioned in your goal: while flexible models like XGBoost or Random Forests can capture non-linearities, they remain sensitive to the training regime8. When a "Black Swan" event occurs that is absent from the training data, even a high-performing model will struggle unless it incorporates broader macroeconomic features beyond the yield curve999.

7 Conclusion

This paper shows that recession forecasting performance is highly regime-dependent. Yield curve factors remain powerful predictors, but their interpretation must account for term premium distortions and structural instability. Combining PCA-based yield representations with macroeconomic indicators and disciplined threshold selection yields robust and interpretable forecasts across diverse economic environments.

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